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Key Players and Environmental Impacts in the Global Plastic Trade Lifecycle

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Abstract

We analysed the global plastic trade network from 1995 to 2021 using a multilayer graph framework, categorizing countries by GDP. The study identified high-income countries as key players across all stages of the plastic lifecycle. In layer 5, focused on plastic waste generation, we used Cross-Correlation, Granger Causality, and VAR analyses to examine interdependencies. Additionally, we investigated the correlation between plastic trade stages and Environmental Performance Index (EPI) indicators. Results show significant historical correlations between different stages of plastic trade but no significant direct correlation with any EPI indicators. This highlights the dominance of high-income countries in plastic trade and the complex relationship between plastic trade and environmental performance, suggesting that the impact on environmental outcomes is not straightforward.

Keywords: plastic trade lifecycle, multilayer, EPI, Cross-correlation.

1. Introduction

The global plastic trade encompasses a comprehensive five-stage life-cycle plastic trade starting from primary plastic forms to plastic waste. While plastic is integral to modern life, it is frequently associated with significant environmental issues, particularly those stemming from the trade of plastic waste. This aspect has received substantial attention, especially following China's 2018 ban on the import of plastic waste. The impact of plastic waste has been studied from various perspectives, including trade dynamics (Bourtsalas et al., 2022; Wang et al., 2020; Zhao et al., 2021, 2022a, 2022b), circular economy frameworks (Gregson & Crang, 2015; Gu et al., 2017; Z. Liu, Adams, & Walker, 2018; Z. Liu, Adams, Cote, et al., 2018; Qu et al., 2019; Wahab & Lim, 2022), and environmental implications (Cotta, 2020; Z. Liu et al., 2021a; Ren et al., 2020, 2023; Tran et al., 2021; Wen et al., 2021). Within the environmental domain, greenhouse gas (GHG) emissions have become a primary metric (Z. Liu et al., 2021a), often analysed through Life Cycle Analysis (LCA) (Gu et al., 2017).

Despite the extensive focus on plastic waste, research addressing the global plastic trade as a whole remains limited. Wang's 2022 study utilized a multilayer network approach to analyse the dynamics of plastic trade, identifying key players in the process (Wang et al., 2022). Barrowclough's 2020 research expanded on this by not only identifying key players but also discussing data generation for each phase of the plastic lifecycle, culminating in a prototype dataset for a plastic trade database (Barrowclough et al., 2020).

However, despite the prevalent association of plastic with environmental issues, comprehensive studies on the environmental impacts of the entire plastic trade lifecycle are scarce. Key areas for further exploration include understanding whether plastic waste is predominantly generated from previous trade activities, identifying major contributors at each stage of the trade across different years, and examining the relationship between plastic trade stages and environmental indicators such as the Environmental Performance Index (EPI).

The EPI is renowned for its comprehensive measurement of environmental performance across 32 indicators (Pandey et al., 2023). Investigating the correlation between each stage of the plastic trade and EPI indicators could yield valuable insights into the environmental impact of global plastic trade. This exploration could inform strategies to mitigate negative environmental effects and promote sustainable practices within the plastic industry. While substantial research has been conducted on plastic waste and its environmental impacts (Pandey et al., 2023), the broader context of the global plastic trade warrants further investigation. Understanding the intricate dynamics of the entire plastic lifecycle, identifying key players, and analysing environmental correlations can provide a more holistic view of plastic's global impact and contribute to the development of more sustainable trade practices.

To comprehensively analyse the global plastic trade network dynamics, we will employ a multilayer network approach. Additionally, to examine the relationship between plastic trade and environmental issues, we will use cross-correlation, Granger causality analysis, and vector autoregression. The results will identify which countries, categorized by income level, play dominant roles in each stage of the plastic trade over the years. Furthermore, we will determine which stages have strong correlations with environmental factors and which stages show direct correlations with specific Environmental Performance Index (EPI) indicators.

The structure of the remaining sections is as follows: Section 2 will highlight the literature review, Section 3 will explain the research methodology, Section 4 will discuss the research findings, and Section 5 will present the conclusion.

2. Literature Review

The global plastic trade involves a complex network of transactions that span multiple stages, from the creation of primary plastic forms to the management of plastic waste. This network has been extensively studied in terms of its economic and industrial dynamics. For instance, Wang in 2022 utilized a multilayer network approach to analyse the dynamics of the global plastic trade, identifying key players and the intricate relationships between different countries involved(Wang et al., 2022). Barrowclough in 2020 expanded on this by creating a prototype dataset that maps the lifecycle of plastic trade, offering a foundational tool for further research(Barrowclough et al., 2020).

The environmental impacts of plastic trade, particularly concerning plastic waste, have been a significant area of concern. The shift in global plastic waste management practices following China's 2018 ban on plastic waste imports has highlighted the need for sustainable practices and policies. Researchers such as Brooks et al. (2018) documented the repercussions of this ban, emphasizing the redistribution of waste exports to other countries and the ensuing environmental consequences (Brooks et al., 2018). Wen et al. (2021) found that this initial shift had negative environmental impacts on global warming but improved other indicators, saving 2.35 billion euros annually in eco-costs globally (Wen et al., 2021). They utilized Life Cycle Analysis (LCA) metrics to reach these conclusions.

Huang in 2020 employed multiregional input-output modelling, structural path analysis (SPA), ecological network analysis (ENA), and a hypothetical extraction method (HEM) to examine the impact of the ban. They discovered that China experienced a shortage of plastic waste post-ban but would have faced issues with unrecyclable waste had the imports continued, with an economic impact of only 0.04% (Huang et al., 2020). Gu et al. (2017), using LCA, found that in plastic recycling, the extrusion process had the most significant environmental impact, with fillers and additives, especially impact modifiers, playing crucial roles (Gu et al., 2017). Ren in 2020 recommended, based on LCA findings, that trading plastic waste is environmentally preferable to recycling it domestically (Ren et al., 2020). Additionally, Geyer in 2017 highlighted the substantial carbon footprint associated with plastic production and disposal, urging the adoption of circular economy principles to mitigate these impacts(Geyer et al., 2017).

The concept of a circular economy has been proposed as a solution to the challenges posed by plastic waste. This approach emphasizes recycling, reusing, and reducing plastic waste to create a sustainable lifecycle for plastic products.

The application of multilayer network analysis in studying trade dynamics offers a robust framework for understanding the complexities of global trade networks. This method allows for examining interdependencies and interactions across different layers of trade activities. Wang in 2022 utilized this approach to identify the key players in the plastic trade and understand the dynamic relationships between them over time (Wang et al., 2022).

Despite extensive research on the economic and industrial aspects of plastic trade, there is a notable gap in studies examining the direct correlation between different stages of plastic trade and environmental performance indicators. The Environmental Performance Index (EPI) provides a comprehensive set of metrics to evaluate environmental performance across various dimensions (Pandey et al., 2023). However, the linkage between plastic trade activities and EPI indicators remains underexplored.

While significant progress has been made in understanding the dynamics of plastic trade and its environmental impacts, critical areas require further investigation. These include analysing the entire lifecycle of plastic trade from plastic raw to plastic waste; identifying the dominant countries in each stage of the plastic trade, categorized by income level; and examining the correlation between plastic trade stages and specific Environmental Performance Index (EPI) indicators. This study aims to address these gaps by employing a multilayer network approach combined with advanced statistical analyses, including cross-correlation, Granger causality analysis, and vector autoregression (VAR), to provide a comprehensive understanding of the global plastic trade network and its environmental implications.

3. Data and Methodology

1. Data

In this research, we utilized several datasets. The first dataset, global plastic trade data, was retrieved from unctad.org, comprising nearly 27 million rows that provide insights into the evolution of global plastic trade over the years. The second dataset is the GDP data from the World Bank, which includes information on all countries worldwide. The third dataset contains Environmental Performance Index (EPI) indicators, sourced from epi.yale.edu, encompassing 32 indicators for over 180 countries globally. Comprehensive data pre-processing was undertaken to ensure all datasets were in the appropriate format for further analysis.

2. Methodology

Once the data had undergone rigorous cleaning processes, the next phase entailed the construction of the network. The network structure adopted for this study is a multilayer directed graph. In this approach, we employed the L-space methodology, where each country represented a node within the network, and trading relationships between countries were depicted as edges (Bai et al., 2023; H. Liu et al., 2018; Teri Aripin & Fuadi, 2023; Xu et al., 2023)..

2.1. Trade Network Construction

We structured the global plastic trade network into five distinct layers within a directed multilayer graph, each layer corresponding to a specific stage of the plastic life cycle. These layers are defined as follows: Layer 1 represents raw plastic material, Layer 2 represents intermediate forms of plastic, Layer 3 represents intermediate manufactured plastic goods, Layer 4 represents finished plastic products, and Layer 5 encapsulates plastic waste.

The construction model employed for this multilayer network is as follows: (Wang et al., 2022)

$$G_{trade} = (G^{y}, C^{y}) \qquad \textit{Equation 1}$$

Where:

G_{trade} = Multilayer directed plastic trade graph

G^{y} = Single layer of plastic trade in certain year

C^{y} = Interconnection between same node in different layer

$$G^{(y,a)} = (N^{(y,a)}, E^{(y,a)}, W^{(y,a)}) \qquad \textit{Equation 2}$$

Where:

$G^{(y,a)}$ = Layer “a” for certain year “y”

$N^{(y,a)}$ = Country (Node) in layer “a” in certain year “y”

$E^{(y,a)}$ = Trade relationship (Edges) in layer “a” in certain year “y”

$W^{(y,a)}$ = Trading volume (Weight) in layer “a” in certain year “y”

$$C^{y} = C^{[y,a,a+1]}, a \in \{1,2,3,4\} \qquad \textit{Equation 3}$$

Where:

C^y = Interconnection between same node in different layer

$C^{[y,a,a+1]}$ = Interconnection in certain layer "a" in certain year "y"

2.2. Network Analysis

We identify key players in the network based on various criteria, including centrality degree, in-degree and out-strength To accomplish this, we calculate the centrality degree for each node in the network, assess both in-degree and out-degree, and extract the countries with the highest trade volume.

Our first objective is to identify the leading plastic trading countries, both as importers and exporters. To achieve this, we employ the weighted centrality degree approach, specifically the strength degree (Bai et al., 2023; Teri Aripin & Fuadi, 2023).

$$W = \sum_{i \neq j} (w_{ij} \text{ or } w_{ji}) \qquad \textit{Equation 4}$$

Where "W" represents the strength of the country and it values is the total trade in thousand metric ton.

After identifying the key players then we characterized those players based on their income. We began our analysis by collecting three main datasets: global plastic trade data, GDP data from the World Bank, and Environmental Performance Index (EPI) indicators. To integrate this information, we merged the global plastic trade data with the Income Group data, using common identifiers such as "Economy Label" or "Partner Label." This merging process allowed us to categorize each country involved in plastic trade according to its income group. Once the data was merged, we grouped it by year, stage of plastic trade (referred to as "layer"), and income group. This grouping enabled us to count the number of countries in each income group participating in each stage of plastic trade for each year, providing a detailed view of participation across different income levels.

2.3. Plastic Waste Generation and EPI analysis

- Cross-Correlation:

In this process, we aim to explore the relationship between different stages of plastic trade by computing the cross-correlation between them. Cross-correlation helps us understand how the values of one time series relate to another time series at different time lags. This means we can see if changes in one stage of plastic trade (e.g., the production phase) are followed by changes in another stage (e.g., the recycling phase) with some delay.

To start, we define a function to compute cross-correlation. Cross-correlation measures the similarity between two time series as a function of the time-lag applied to one of them. Mathematically, the cross-correlation function (CCF) at lag $k$for two time series X and Y is defined as:

$$CCF_{XY}(k) = \frac{1}{N-k} \sum_{i=1}^{N-k} \frac{(X_t - \mu_X)(Y_{t+k} - \mu_Y)}{\sigma_X \sigma_Y} \quad \text{Equation 5}$$

Where:

- X_t and Y_t are the values of the time series X and Y at time t.
- μ_X and μ_Y are the means of X and Y,
- σ_X and σ_Y are the standard deviations of X and Y,
- N is the number of observations in the time series,
- k is the lag, which can be positive or negative,

Using this function, we compute the cross-correlation values for different lags, ranging from -10 to +10. A positive lag means we are checking how past values of one series relate to current values of another series, while a negative lag means we are checking how future values of one series relate to current values of another series.

Next, we apply this function to compute the cross-correlation between the fifth stage of plastic trade (layer 5) and each of the other stages (layers 1 through 4). We then plot these cross-correlation values against the different lags. This plot allows us to visually inspect the relationship between layer 5 and the other layers at various time lags.

In the plot:

- The x-axis represents the lag, ranging from -10 to +10.
- The y-axis represents the cross-correlation values.
- Each line in the plot corresponds to a different stage of plastic trade, showing how it correlates with layer 5 at different lags.

By analysing this plot, we can identify if there are any significant correlations at specific lags, indicating potential lead-lag relationships between different stages of plastic trade. This insight can help us understand the temporal dynamics and interdependencies within the plastic trade lifecycle.

- Granger Causality Test

To determine if changes in different stages of the plastic trade (referred to as layers 1, 2, 3, and 4) can help predict changes in the final stage (layer 5), we utilize the Granger causality test. This statistical test examines whether past values of one time series contain information that can predict another time series beyond what the past values of the second time series can predict on their own.

In our analysis, we perform Granger causality tests for each layer (layer 1, layer 2, layer 3, and layer 4) to determine if they can predict layer 5. We use a maximum lag of 5 periods for the test.

F-Statistics and P-Values: For each lag, the F-statistic measures the overall significance of the lagged values of X. The p-value associated with this F-statistic helps determine if the results are statistically significant. A low p-value (typically less than 0.05) indicates that the null hypothesis can be rejected, suggesting that X Granger-causes Y.

- Vector Autoregression

To analyse the dynamic relationships between the different stages of the plastic trade, we employ a Vector Autoregression (VAR) model. This statistical model is used to capture the linear interdependencies among multiple time series. A VAR model consists of a system of equations where each variable is a linear function of its own past values and the past values of all other variables in the system. By using the VAR model, we can effectively analyse the interconnected dynamics of the plastic trade stages and forecast their future behaviour based on past trends. This helps in understanding the temporal relationships and potential future patterns in the global plastic trade.

- Pearson correlation coefficient

In the last analysis, we aim to uncover the relationship between different stages of the global plastic trade and various Environmental Performance Index (EPI) indicators. Our approach involves several key steps to ensure that the data from these two sources are comparable and that we can accurately calculate the correlations.

First, we identify the years that are common between the plastic trade data and the EPI data, allowing us to focus on a consistent time frame for our analysis. We then filter both datasets to include only these common years, ensuring alignment in the temporal scope. To maintain consistency, we standardize the column names in the plastic trade data, making sure that the country names match those in the EPI data. This harmonization is crucial for merging the datasets later.

Next, we reshape the plastic trade data into a wide format. In this format, each row represents a unique combination of country and year, with separate columns for each stage of the plastic trade, referred to as layers. This restructuring allows us to easily compare the plastic trade metrics with the EPI indicators.

We then merge the filtered EPI data with the reshaped plastic trade data, resulting in a combined dataset that includes both environmental performance indicators and plastic trade metrics for each country and year. This comprehensive dataset sets the stage for our correlation analysis.

To identify the relationship between the plastic trade stages and EPI indicators, we loop through each EPI indicator and calculate its correlation with each stage of the plastic trade. For each EPI indicator, we filter out rows with missing values to ensure the accuracy of our calculations. We then compute the correlation, which measures the strength and direction of the linear relationship between the EPI indicator and the plastic trade stage.

Correlation is typically quantified using the Pearson correlation coefficient, which ranges from -1 to 1. A value of 1 indicates a perfect positive linear relationship, -1 indicates a perfect negative linear relationship, and 0 indicates no linear relationship.

Finally, we store the correlation results in a structured format and convert them into a tabular form for better visualization and interpretation.

4. Result and Discussion

- Global plastic trade network

The global plastic trade network comprises five distinct layers, each representing a different stage in the plastic product lifecycle. In this network, Layer 1 signifies the trade in primary forms of plastic, while Layer 2 represents intermediate forms of plastic. Moving further along, Layer 3 deals with intermediate manufactured plastic goods, Layer 4 encompasses the trade in finished plastic products, and Layer 5 focuses on the exchange of plastic waste.

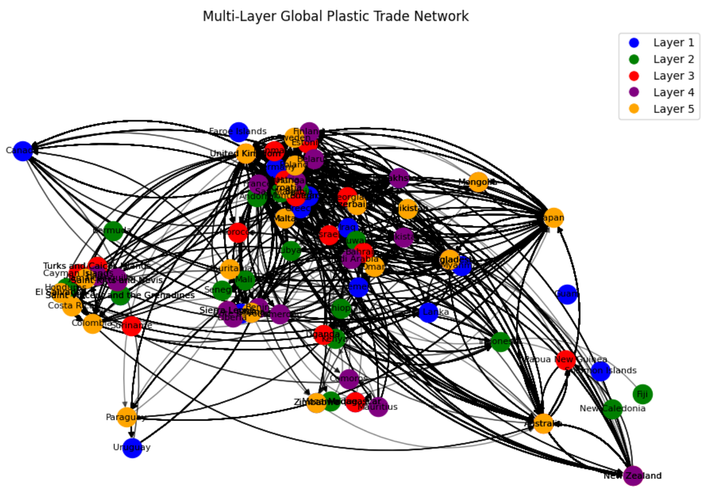

Figure 1. Illustration of Multilayer Global Plastic Trade Network. Due to space limitation we only show 20 countries for each layer.[1]

Figure 1 provides a visual representation of this intricate network. Due to the substantial volume of data, we've chosen to display a limited subset of 20 nodes for each layer, offering a snapshot of the network's complexity and connectivity within these different plastic trade stages.

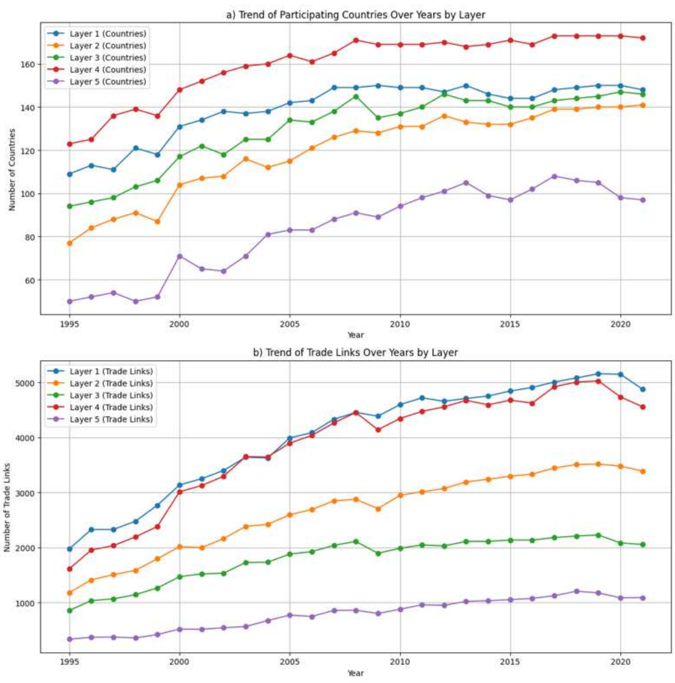

Figure 2. Trend of Global Plastic Trade Network Growth.[2]

[1] In building the global plastic trade network visualization we implemented equation 1 and equation2.
[2] We implement equation 1 – 3 to come up with that graph.

The continuous increase in both the number of nodes and edges in the global plastic trade network, as seen in Figure 2a and Figure 2b, signifies significant growth and activity within the industry over two decades. Notably, the doubling of nodes in Layer 5 suggests the entry of new players and greater diversity in participants, reflecting a broader global market and more complex network, it is in line with what has been discussed by Liu et al (2021) that global plastic waste trade (GPWT) grow that somehow it increase the GHG emission caused by its transportation (Z. Liu et al., 2021b). Furthermore, the tripling of edges in Layer 1 and Layer 4 indicates enhanced connectivity and a more intricate supply chain.

To gain deeper insights into trade behavior, we opted to dissect the network into two distinct segments: exports and imports. This division provides a more nuanced understanding of how plastic trade operates. The trend in overlapping nodes between these exports and import networks mirrors what's depicted in Figure 3 and Figure 4.

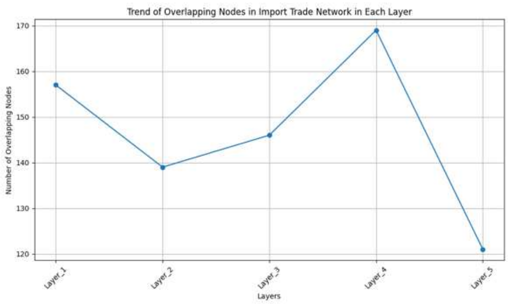

Figure 3. Overlapping countries in different stages as importers in the global plastic trade.

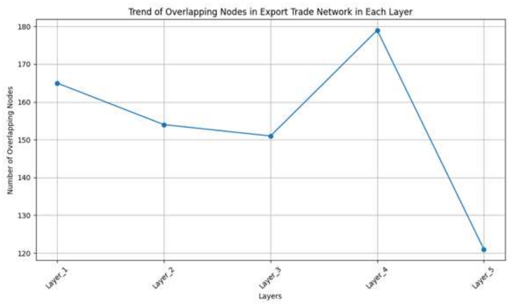

Figure 4. Overlapping countries in different stages as exporters in the global plastic trade

It's intriguing to observe a decline in overlapping nodes from Layer 1 to Layer 2[3]. This suggests that, even as the number of nodes and edges grows, many newcomers are unique to their respective layers, indicating distinct trade patterns. Moreover, there's a notable upsurge in Layer 4, indicating a substantial number of nodes engaged in the trade of finished plastic products. Conversely, Layer 5 experiences a pronounced decline, implying that only a limited number of nodes are involved in plastic waste

[3] We implement equation 3 for calculating overlapping nodes.

trading. These trends shed light on the evolving dynamics within the plastic trade network, revealing shifts in focus and specialization across different layers of the industry. This finding is similar to what have been discussed by Wang et al (2022) stating that plastic trade has unique character for each layer (Wang et al., 2022).

- Players Grouped by Income Level

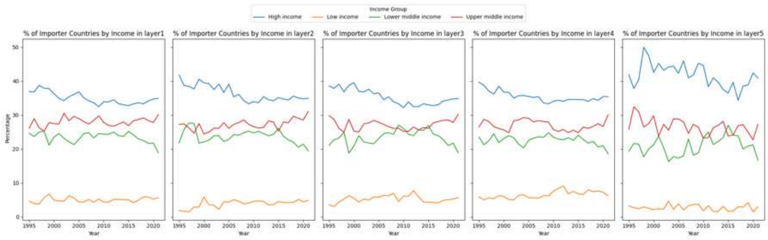

Figure 5. Percentage Importers Grouped by Income Level

Figure 5 illustrates the percentage distribution of importer countries by income groups across different stages of the plastic trade network from 1995 to 2020. High income countries consistently dominate importer participation in all layers, as evidenced by the blue line, which remains the highest throughout the years, fluctuating between 35% to 45%. Upper middle income countries, represented by the red line, are the second largest group of importers, showing stable participation with percentages generally ranging from 20% to 30%. Lower middle income countries, indicated by the green line, also maintain a steady presence, typically ranging between 15% to 25%. In contrast, low income countries, represented by the orange line, consistently show the lowest participation, hovering around 5% to 10%. Across all layers, the relative ranking of the income groups remains consistent, with high income countries leading, followed by upper middle income, lower middle income, and low income countries.

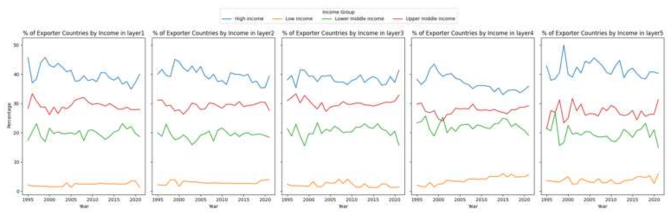

Figure 6. Percentage Exporters Grouped by Income Level

Similar to importers, Figure 6 illustrates the distribution of exporter countries in the global plastic trade. The trend closely mirrors that of importers, with high-income countries dominating the trade, while low-income countries remain a minority in trade activities. This consistent pattern indicates that economic capacity significantly influences a country's participation in the global plastic trade network, with high-income countries being the most involved in both importing and exporting activities. Overall, the data underscores the substantial role of economic status in determining a country's engagement in the global plastic trade.

- Waste Generation and EPI Analysis

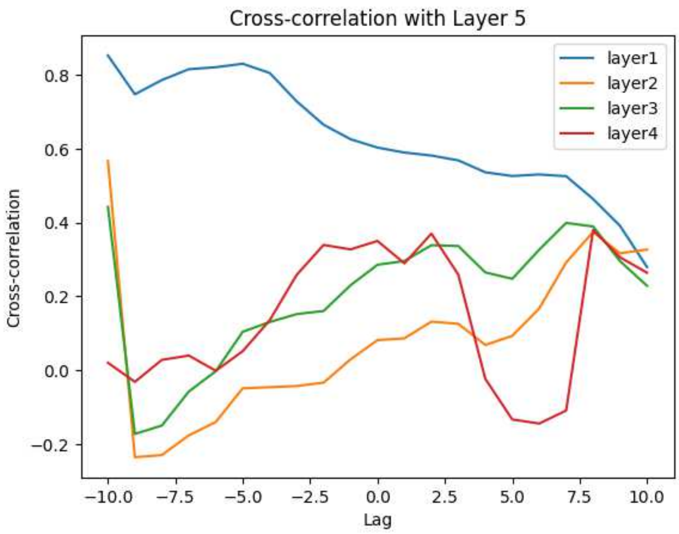

Figure 7. Cross-Correlation Analysis Result for Plastic Waste Generation. It shows trading volume in layer 1 have the strongest correlation with layer 5[4].

The cross-correlation analysis shown in Figure 7 reveals a strong relationship between trade volumes in `layer1` and plastic waste generation in `layer5`. Specifically, `layer1` shows the strongest correlation with `layer5`, particularly at a lag of -10, with a cross-

[4] This analysis employed equation 5

correlation coefficient close to 0.8. This indicates that past trade volumes in `layer1` are strongly related to current plastic waste generation. In other words, higher trade volumes in the early stages of plastic trade (layer1) have a significant and immediate impact on the amount of plastic waste generated later (layer5).

In contrast, the correlations between `layer5` and the other layers (2, 3, and 4) are weaker. For instance, `layer2` shows a high correlation at negative lags, which diminishes for positive lags, indicating a less direct relationship. `Layer3` exhibits a moderate correlation with a peak around positive lags 7-8, suggesting a delayed relationship between trade volumes in `layer3` and plastic waste generation. Similarly, `layer4` displays an increasing correlation towards positive lags, implying a moderate predictive relationship.

The Granger causality analysis investigates whether past values of one time series can be used to predict another, revealing significant insights. Specifically, `layer4` exhibits substantial Granger causality with `layer5` at higher lags. For example, at lag 2, the ssr based chi2 test produces a chi2 value of 7.9065 with a p-value of 0.0192, while the likelihood ratio test yields a chi2 value of 6.8699 with a p-value of 0.0322. As the lag increases to 3, the ssr based chi2 test results in a chi2 value of 21.4437 with a p-value of 0.0001, and the likelihood ratio test gives a chi2 value of 15.3221 with a p-value of 0.0016. At lag 4, these values escalate significantly, with the ssr based chi2 test showing a chi2 value of 70.8688 and a p-value of 0.0000, and the likelihood ratio test producing a chi2 value of 32.3473 with a p-value of 0.0000. At lag 5, the ssr based chi2 test results in a chi2 value of 81.4623 with a p-value of 0.0000, and the likelihood ratio test yields a chi2 value of 34.0596 with a p-value of 0.0000. These results strongly suggest that past values of `layer4` are predictive of future values of `layer5`.

Additionally, the Vector Autoregression (VAR) model captures the linear interdependencies among multiple time series, providing further insights. The analysis shows that layers 1 to 4 are highly interrelated, with significant correlations among themselves. For instance, the correlation coefficient between `layer1` and `layer2` is 0.860514, and between `layer2` and `layer3` is 0.948303, indicating strong interdependencies. However, `layer5` does not exhibit strong correlations with layers 1 to 4 within the model. For example, the correlation coefficient between `layer1` and `layer5` is -0.128209, signifying a weak relationship. These findings highlight the complex dynamics within the plastic trade network and emphasize the significant role of `layer4` in predicting plastic waste generation in `layer5`.

Overall, the analysis reveals key insights into the dynamics of the plastic trade network. First, the cross-correlation analysis shows a strong historical influence of `layer1` on `layer5`, with a high cross-correlation coefficient of around 0.8 at a lag of -10, indicating that past trade volumes in `layer1` are significantly related to current plastic waste generation. Second, the Granger causality tests highlight the predictive power of `layer4` for `layer5`, with strong statistical significance at higher lags, such as a chi2 value of 70.8688 and a p-value of 0.0000 at lag 4, suggesting that past values of `layer4` can effectively predict future plastic waste generation. Lastly, the VAR model demonstrates strong interrelationships among layers 1 to 4, evidenced by high correlation coefficients like 0.860514 between `layer1` and `layer2`, and 0.948303 between `layer2` and `layer3. However, `layer5` does not exhibit strong correlations with these layers, indicating a more complex relationship with the rest of the network.

Table 1. Pearson Correlation Coefficient between EPIs Indicators and Layers in Global Plastic Trade. 1 being perfect positive correlation, -1 being negative correlation and 0 suggest no correlation.

Correlation between Plastic Trade Layers and EPI Indicators

EPI Indicators	layer1	layer2	layer3	layer4	layer5
BCI	-0.18	-0.079	-0.11	-0.037	-0.12
CDI	0.11	0.06	0.11	0.032	0.15
CEH	0.03	0.034	0.074	0.012	0.065
CHI	-0.19	-0.084	-0.12	-0.035	-0.14
DBI	-0.02	-0.029	-0.014	-0.0044	-0.044
DCI	0.00092	-0.011	-0.0076	0.0018	-0.015
DMI	0.0083	-0.0069	-0.051	0.0079	-0.073
DNI	-0.0063	-0.013	-0.054	0.0096	-0.08
DSI	-0.061	-0.085	-0.059	-0.051	-0.057
DXI	-0.04	-0.054	-0.031	-0.026	-0.042
FSS	0.063	0.021	0.02	0.0019	0.014
HAD	-0.15	-0.14	-0.11	-0.12	-0.098
MPA	0.29	0.14	0.19	0.079	0.24
MTR	0.016	-0.0063	0.0039	-0.013	0.014
NOI	-0.13	-0.056	-0.083	-0.028	-0.093
NXI	-0.14	-0.056	-0.069	-0.016	-0.079
PAR	-0.052	-0.037	-0.066	-0.032	-0.044
PBD	-0.12	-0.12	-0.058	-0.083	-0.07
PME	0.12	0.15	0.22	0.019	0.23
PMW	0.079	0.11	0.2	0.0044	0.21
SDI	-0.076	-0.031	-0.03	-0.007	-0.032
SHI	-0.0089	0.0045	-0.017	-0.0093	-0.041
SNM	-0.27	-0.28	-0.22	-0.25	-0.23
SPI	0.076	0.034	0.045	0.003	0.049
TBG	0.078	0.044	0.067	0.015	0.082
TBN	0.05	0.028	0.035	0.0054	0.041
TCL	-0.0089	-0.0029	-0.0066	-0.00073	-0.0096
TPI	-0.0038	-0.0059	-0.0041	-0.00044	-0.0036
USD	-0.12	-0.11	-0.1	-0.1	-0.099
UWD	-0.12	-0.11	-0.11	-0.1	-0.1
WWT	0.2	0.21	0.092	0.19	0.13

Plastic Trade Layers

The correlations between each stage of plastic trade and various Environment Performance Index (EPI) indicators generally show weak relationships. Correlation coefficients, which range from -1 to 1, indicate the strength and direction of a linear relationship between two variables. A coefficient of 1 signifies a perfect positive correlation, -1 indicates a perfect negative correlation, and 0 suggests no correlation. In this analysis which is shown in Table 1, most correlation coefficients are close to 0, suggesting that changes in plastic trade volumes do not significantly influence changes in EPI indicators or vice versa.

However, some positive correlations are noteworthy. For instance, Marine Protected Areas (MPA) exhibit moderate positive correlations with layer 1 (0.288) and layer 5 (0.241), implying a possible association between higher plastic trade and better marine protection efforts, though causality cannot be established from correlation alone.

Similarly, Wastewater Treatment (WWT) shows moderate positive correlations, particularly with layer 2 (0.208) and layer 1 (0.195), indicating that countries with higher plastic trade might also invest more in wastewater treatment.

On the other hand, certain indicators display moderate negative correlations. Soil Nitrogen Management (SNM) shows the strongest negative correlation with layer 2 (-0.278), suggesting that higher plastic trade might be linked to poorer soil nitrogen management practices. Household Air Quality (HAD) and the Child Health Index (CHI) also show notable negative correlations, indicating potential adverse effects of increased plastic trade on these indicators.

Most other indicators, such as Biodiversity Conservation Index (BCI), Climate and Disaster Index (CDI), and Chemical Exposure Health (CEH), exhibit correlations close to zero, suggesting no significant relationship with plastic trade stages. This implies that plastic trade volumes do not have a substantial impact on these environmental performance indicators.

Overall, the weak correlations suggest no strong linear relationship between the stages of plastic trade and the EPI indicators. This indicates that the volume of plastic trade does not significantly affect environmental performance as measured by the EPI indicators. It is crucial to note that correlation does not imply causation, and further in-depth analysis would be necessary to understand any underlying causal relationships.

5. Conclusion

This research comprehensively analyses the global plastic trade, examining its dynamics and environmental impacts through various stages, from primary plastic forms to plastic waste. High-income countries consistently dominate both importing and exporting activities across all stages, highlighting the significant influence of economic capacity on participation in the global plastic trade.

A strong historical influence of initial trade volumes (`layer1`) on subsequent plastic waste generation (`layer5`) was identified, with a notable cross-correlation coefficient close to 0.8 at a lag of -10. Granger causality analysis further revealed that past values of the manufacturing and processing stage (`layer4`) can predict future values of plastic waste generation (`layer5`). The Vector Autoregression (VAR) model showed strong interrelationships among layers 1 to 4 but a weaker relationship between these layers and plastic waste generation (`layer5`).

Correlations between plastic trade stages and Environmental Performance Index (EPI) indicators were generally weak, indicating that changes in plastic trade volumes do not significantly influence environmental performance. However, some positive correlations suggest that higher plastic trade volumes might be associated with better marine protection efforts and wastewater treatment investments, while certain negative correlations imply potential adverse environmental effects on soil nitrogen management and household air quality.

Overall, the research underscores the complexity of the global plastic trade and its nuanced relationship with environmental performance, emphasizing the need for further investigation into causative relationships and the development of sustainable trade practices to mitigate negative environmental impacts.

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